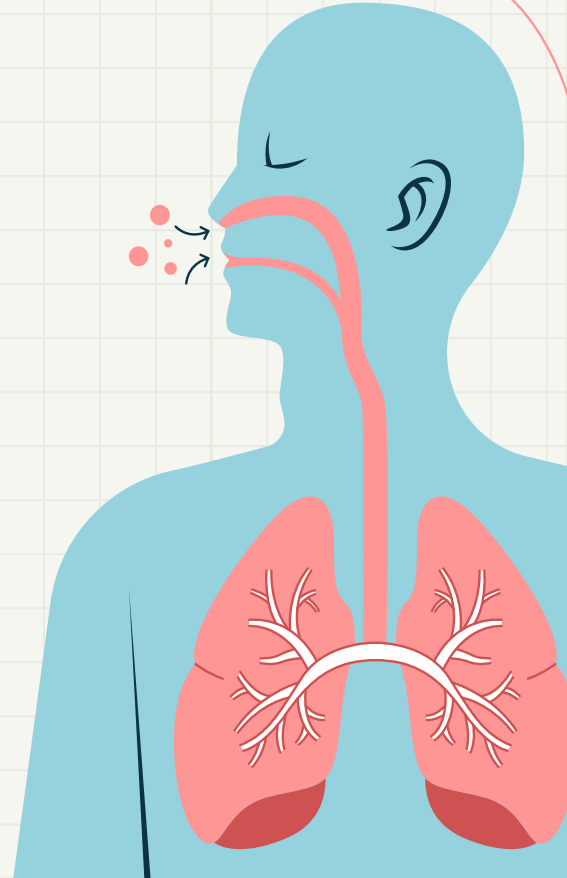


Compression Shirt MEG

By Allen, Anahy, Jeannette, Lisbeth,
Zaid, Alice, and Ira





Outline

01

Background

Motivation behind our project

02

MEG Application

MEG function

03

Design

Materials

04

Testing Protocol

Testing setup

05

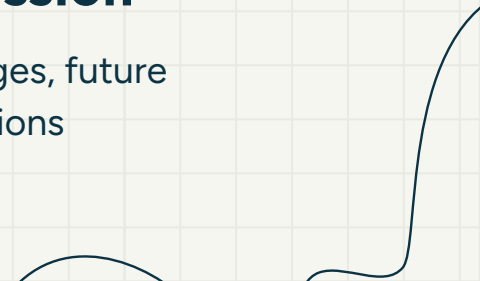
Data and Results

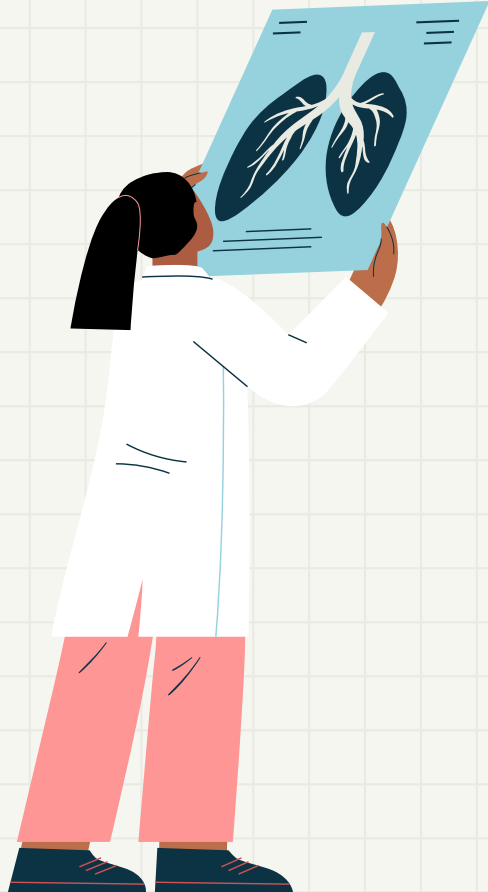
Graphs and analysis

06

Discussion

Challenges, future applications





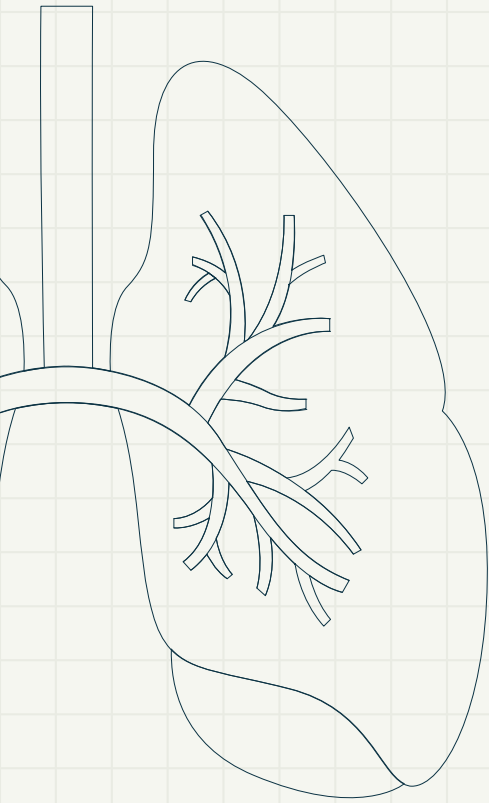
01

BACKGROUND

Why did we do this?

- Respirations are one of the main vital signs acquired for diagnoses
- Can detect various health disorders
 - Asthma
 - COPD
 - Respiratory infections
- Can be commercial usages
 - Athletics
 - Stress & Anxiety Management
- Wearable device market is growing.



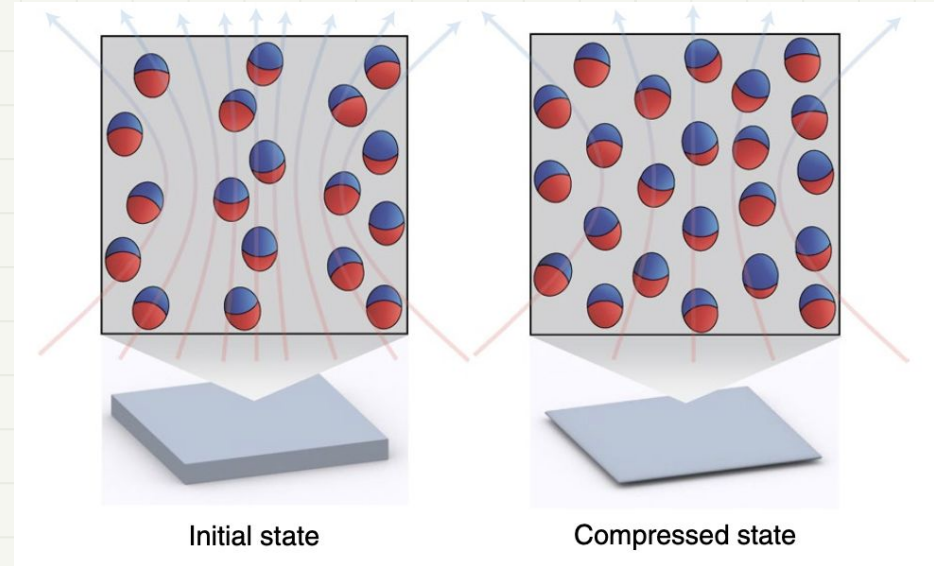


02

MEG APPLICATION

MEG WORKING PRINCIPLE

- Relies on **two** main principles:
 - Magnetoelastic effect
 - Magnetic properties of a material change when subjected to mechanical stress
 - Electromagnetic induction
 - The production of current as a result of changing magnetic fields through a solenoid
- Magnetoelastic effect in soft systems
 - Small pressure yields high electrical output as magnetic nanoparticles are dispersed in a polymer matrix
 - MDDI, MPPI



OUR APPLICATION



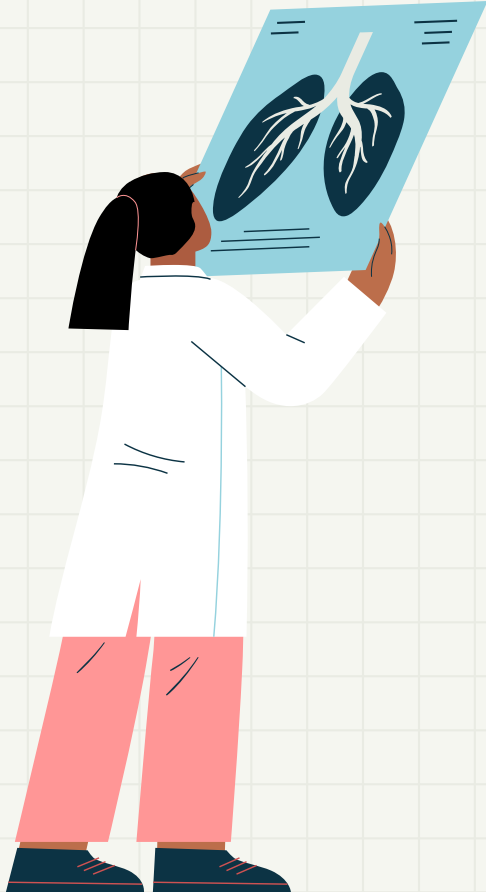
- A compression shirt to track respiration patterns

Main applications:

- Health applications: to identify irregularities and diagnose respiratory
- Sport application: to ensure athletes maintain an optimal breathing rate for their sport activity

Why MEG?

- Waterproof, flexible, simple structure, durable compared to TENG



03

DESIGN

MEG COMPRESSION SHIRT



MATERIALS :

Shirt - 85% Polyester, 15% Polyurethane

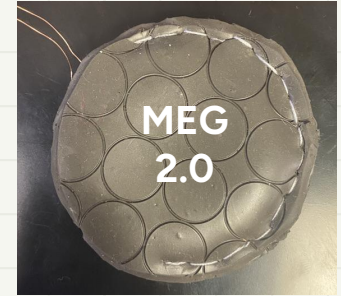
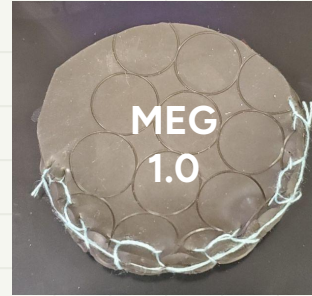
Thread : Cotton

MEG 1.0 - 10.70 g of Ecoflex, 10.80 of Magnetic powder

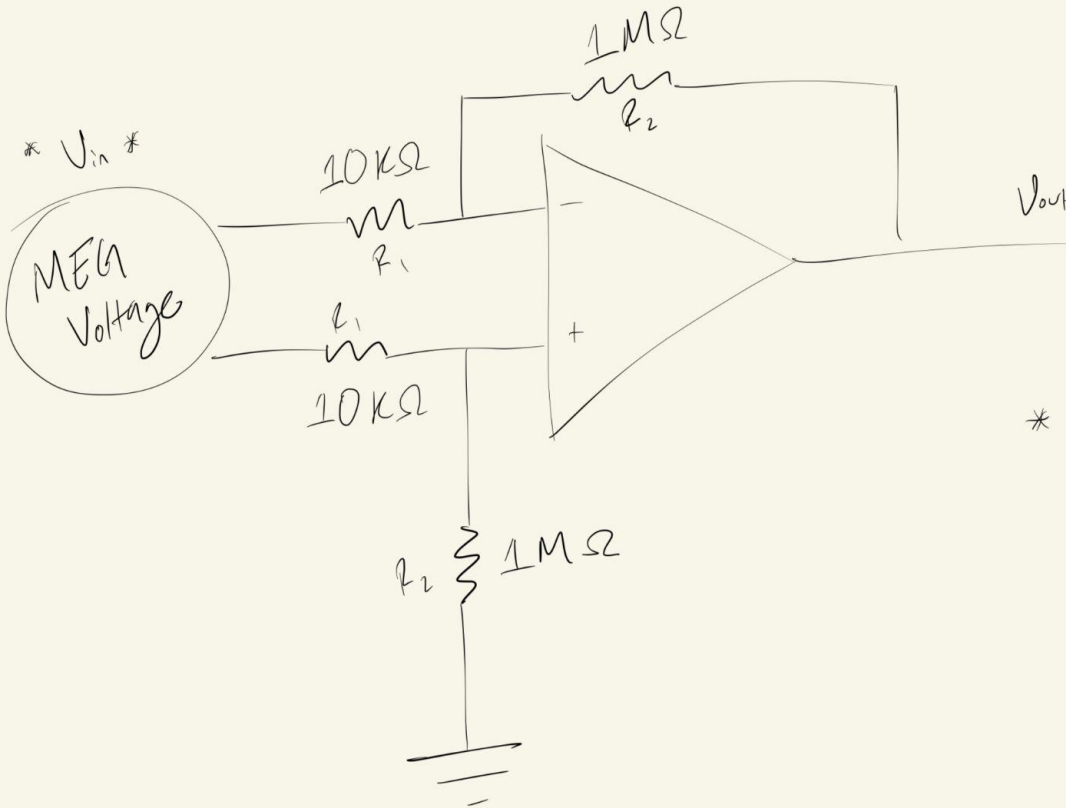
MEG 2.0 - 20.22 g of Ecoflex, 40.270 of Magnetic powder

Conductive coil : 100 turns of Copper Wire

DESIGN :



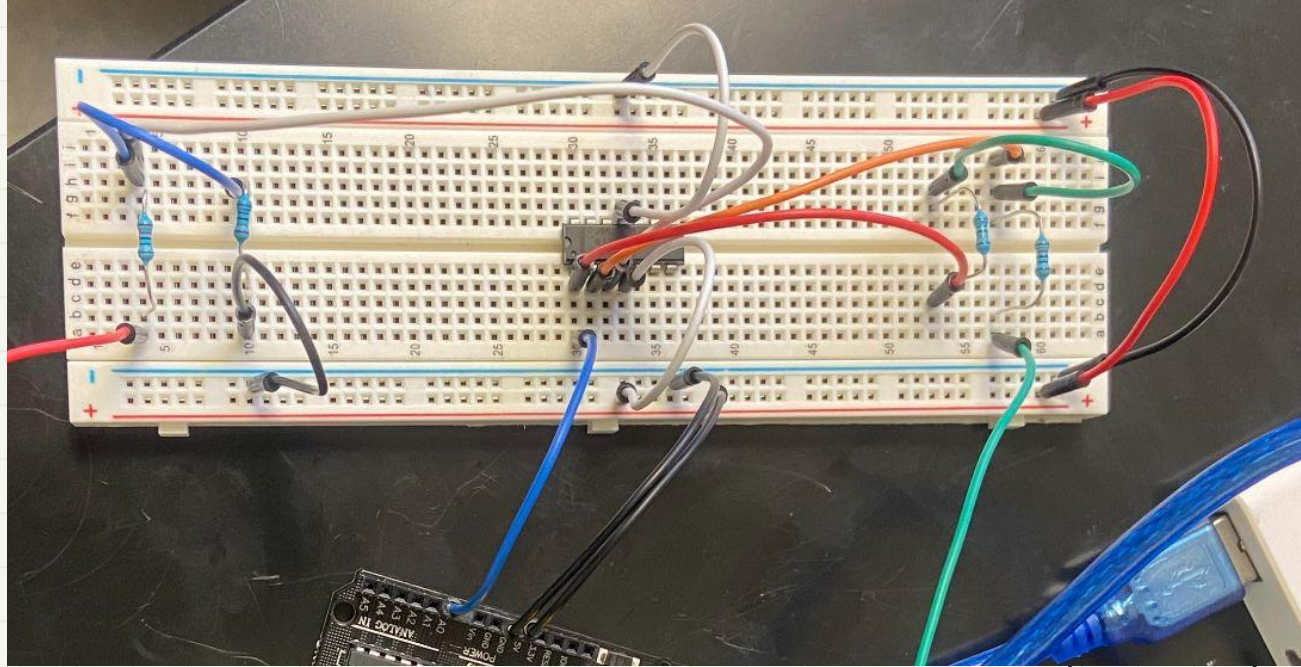
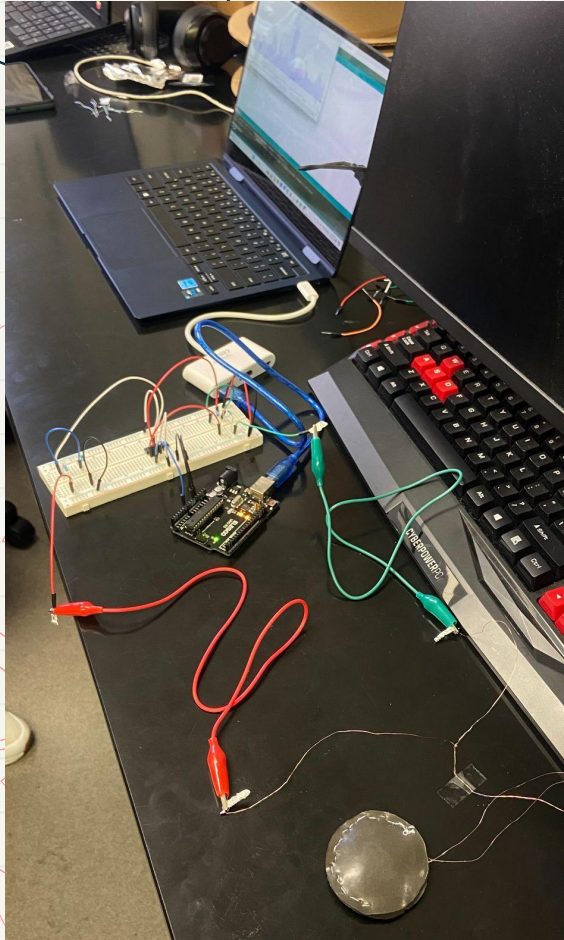
CIRCUIT

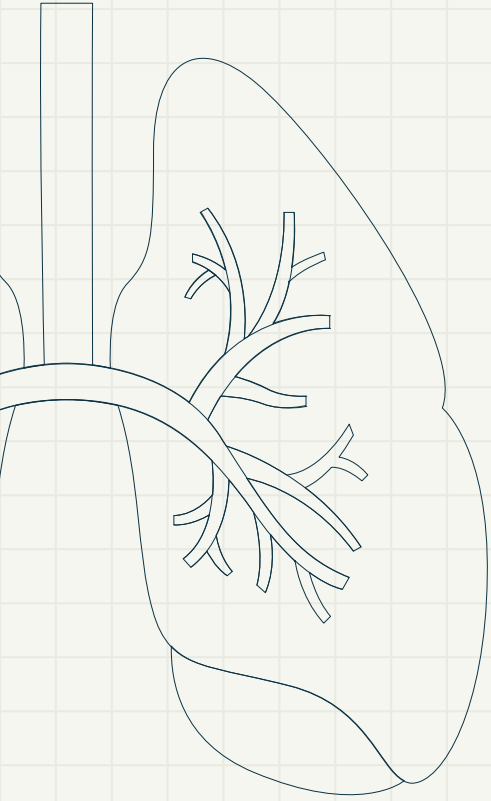


$$V_{out} = (V_{in}) \left(\frac{R_2}{R_1} \right)$$
$$= (V_{in})(100)$$

* V_{out} was fed to *
the arduino

CIRCUIT





04

TESTING PROTOCOL



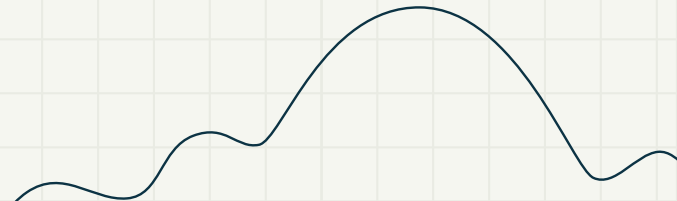
QUALIFICATIONS

- Make sure the shirt fit snug
- MEG was centered on the diaphragm



TESTING

- Using the wires at the bottom of the shirt we connected them to the machine and took the currents



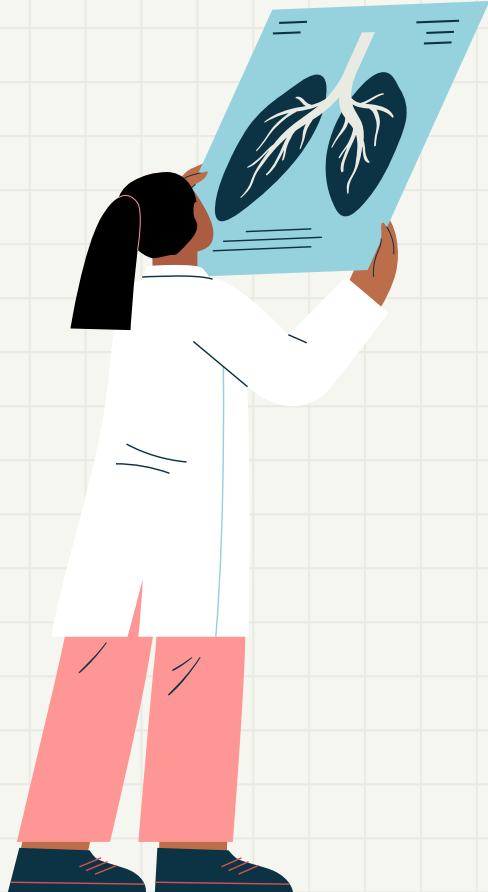


TESTING

- We took data from rapid breathing, deep breaths, and normal breathing

- To record data to monitor breathing after exercise we had Zaid run to get his breathing up
- Immediately after coming back into the lab we recorded his data.



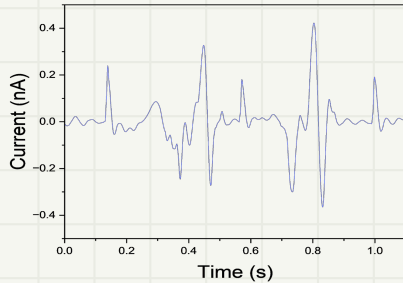


05

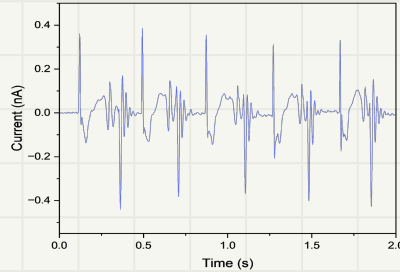
DATA & RESULTS

Test Day #1

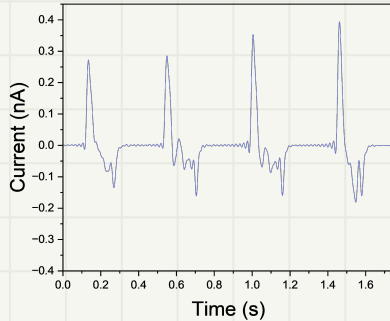
MEG 1.0 - First we conducted test trials only on the MEG's to compare the current output using different turns of copper wire.



25 (smaller) coil turns

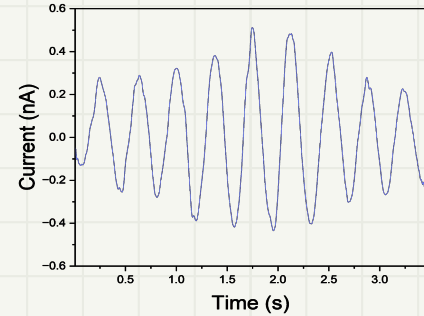


50 (smaller) coil turns



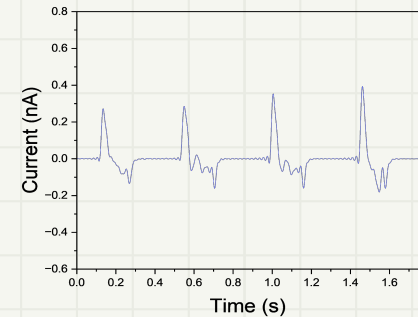
100 (smaller) coil turns

Then we tested the MEG Device on Zaid incorporating only the 50 and 100 turns of copper wire



50 smaller coil turns

(Zaid's respiration after running + breathing hard)



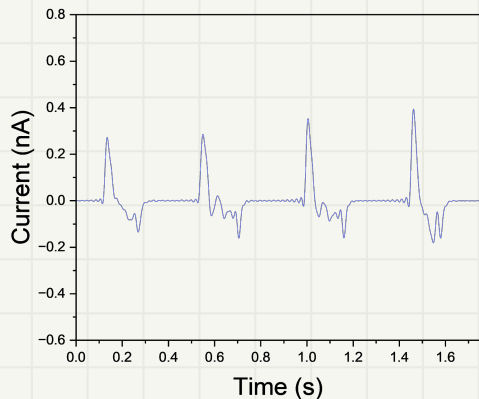
100 smaller coil turns

(Zaid's respiration after running + breathing hard)

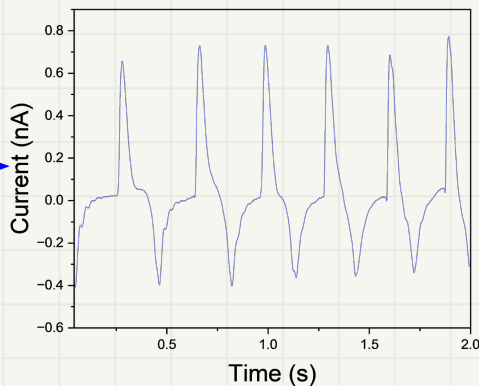
Test Day #2

MEG 1.0 with 100 (larger) coil turns

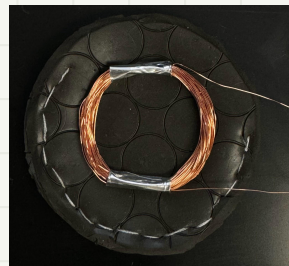
Made the switch to using **larger coil turns** for our testing since we noted that there was an overall **stronger current output**.



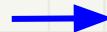
100 **smaller** coil turns



100 **larger** coil turns



smaller coil turns
compared to MEG size



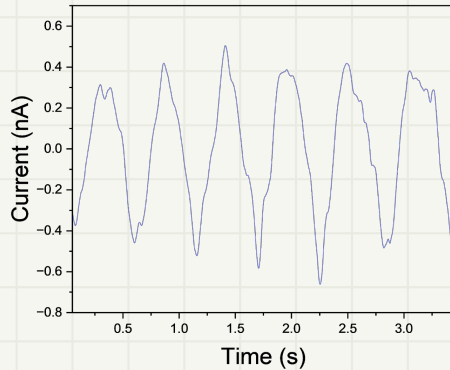
larger coil turns
compared to MEG size

Test Day #2 (cont'd)

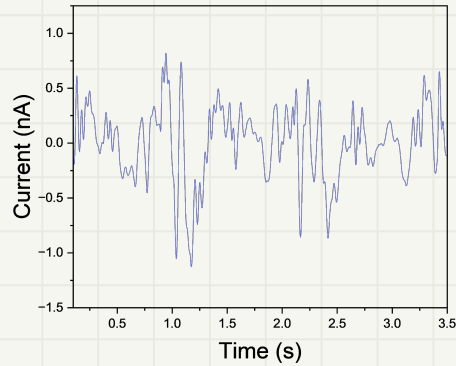
MEG 1.0 with 100 (larger) coil turns

Decided to test on 4 other groupmates to see if we could get stronger signal outputs for respiration monitoring

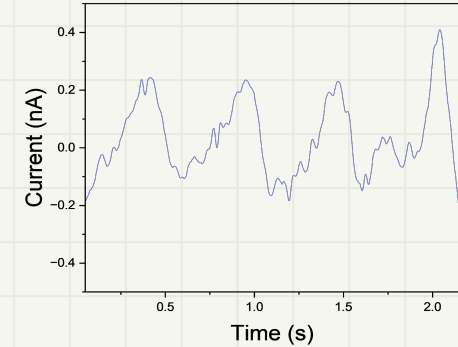
***Note: All group members were standing**



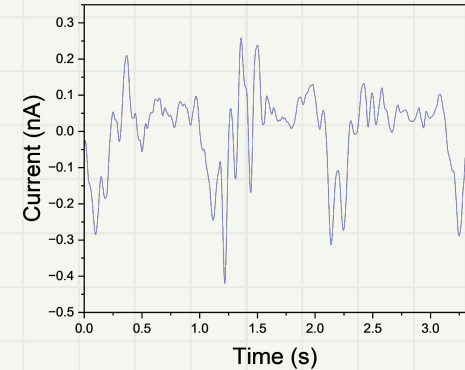
100 larger coil turns - (Allen)



100 larger coil turns - (Jean)



100 larger coil turns - (Lisbeth)

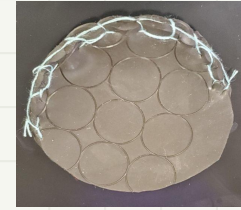


100 larger coil turns - (Anahy)

Test Day #3

MEG 2.0 with 100 (larger) coil turns

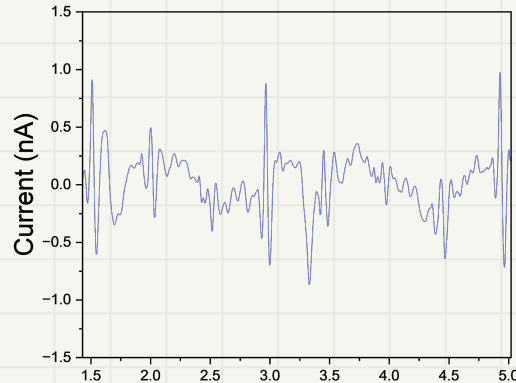
Using **larger coil turns and a new MEG** for our testing we monitored respiratory patterns of only one of our group members, Zaid. The analysis of his results is seen below:



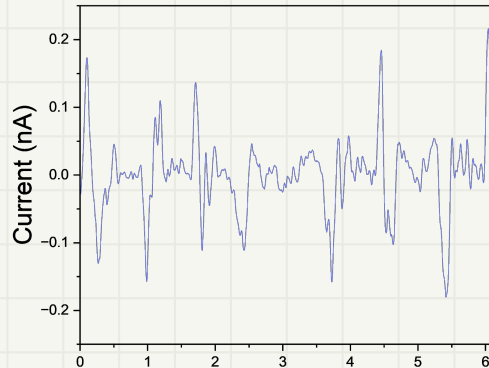
MEG 1.0
1:1 Ratio of
Ecoflex to
Magnetic Powder



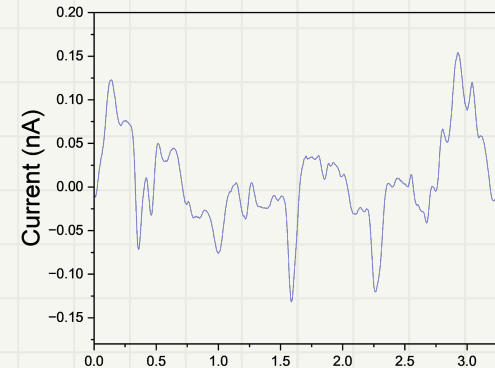
MEG 2.0
1:2 Ratio of Ecoflex
to Magnetic
Powder



(Breathing pattern after running)



(Deep breathing)

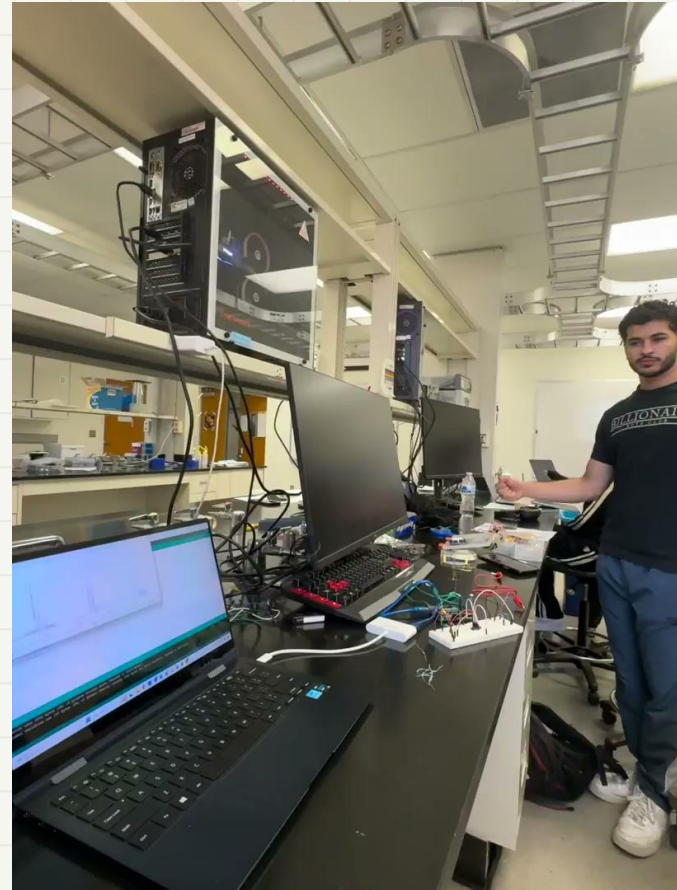
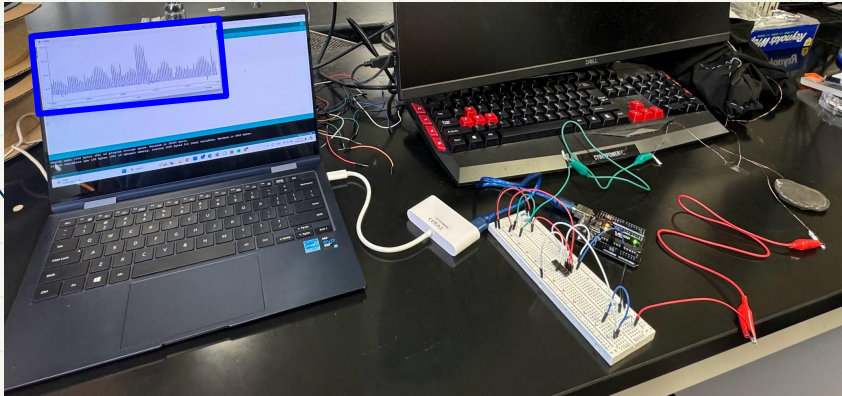


(Normal breathing)

Test Day #3 (Arduino)

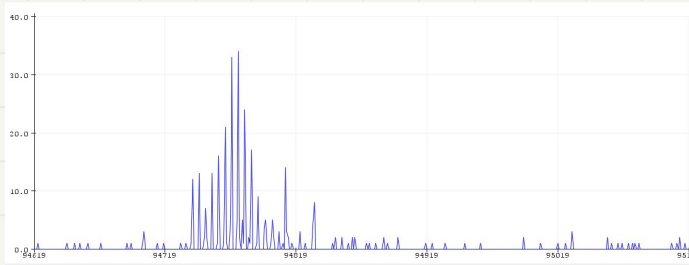
MEG 2.0 with 100 (larger) coil turns

Connected our MEG device to an Arduino Breadboard to obtain signal outputs on our laptop

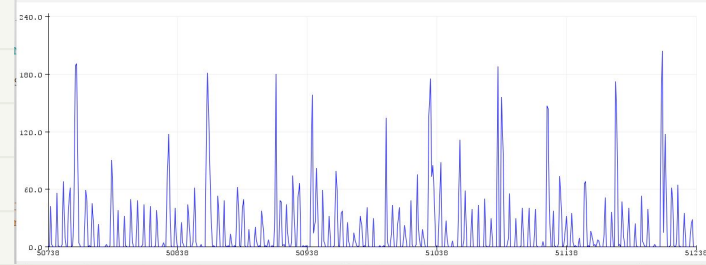


Test Day #3 (Arduino cont'd)

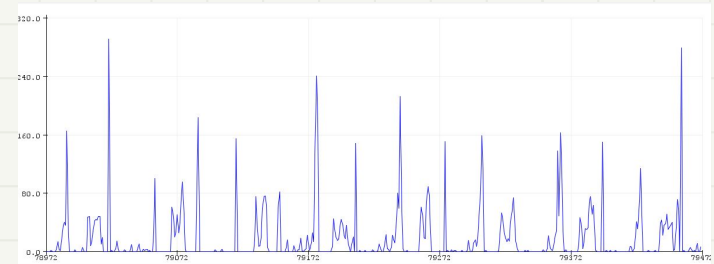
MEG 2.0 with 100 (larger) coil turns



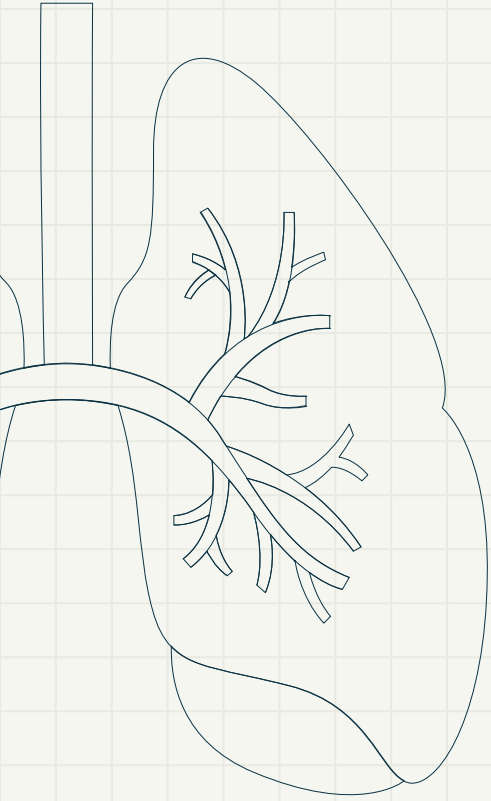
MEG Device monitoring slow
breathing patterns



MEG Device monitoring deep
breathing patterns



MEG Device monitoring rapid
breathing patterns



06

DISCUSSION

Challenges

Wire Placement

- Ran wires down the side seam encased by fabric so that it's out of the way

Sensitivity

- Increased number of coils for copper wire from 50 to 100 coils with larger coil turns

Incorporating MEG

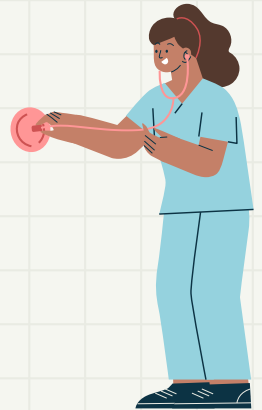
- Used thin MC layer and wire to be able to sew into shirt

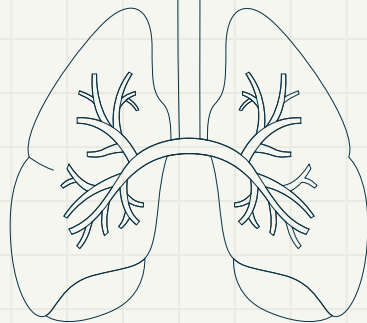
Bluetooth Connectivity

- Bluetooth chip would not connect to any device or disconnect randomly
- Used physical ammeter in lab instead

Future Directions

- Bluetooth
 - Better bluetooth chip and arduino components
- User Interface
 - Create an interactable interface for users to analyze their own respiration
- Adding heart rate monitoring
 - Can pick up on heart beat using current MEG shirt to monitor heart rate
- MEG Textile
 - Integrate MEG textile into shirt instead of sewing soft MEG into shirt





THANK YOU!

To the TA's Xiao and Joyce, BE 167L Class,
and Professor Chen

